
ChronosLOB: Limit Order Book Forecasting with Execution-Aware Validation

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Abstract

Limit order book forecasting is commonly evaluated through classification scores, although the usefulness of a directional signal also depends on confidence, coverage and execution sensitivity. We introduce ChronosLOB, an empirical framework that separates these objects through chronological evaluation, training-only preprocessing, matched representation comparisons and execution-aware diagnostics. A 180-cell FI-2010 benchmark compares matrix transformers and a DeepLOB-style reference across five folds, three seeds, three lookbacks and two horizons. The transformer achieves macro-F1 0.6013 ± 0.1950 , against 0.5133 ± 0.0670 for the reference, but its advantage reverses at the longest lookback. Across 30 matched pairs, market-state SSL-v2 changes macro-F1 by $+0.0112$, MCC by $+0.0259$, ECE by -0.0030 and Brier score by -0.0235 ; improvements concentrate at horizon 10, with deterioration at horizon 50. Snapshot order-flow proxy removal produces the largest feature-ablation effect, with macro-F1 deteriorating in all 100 matched feature-removal comparisons. In a separate diagnostic cohort, confidence filtering sharply contracts directional coverage, while predictive quality varies non-monotonically and cost and latency stresses expose additional sensitivity. These results show that configuration sensitivity and selective coverage are necessary companions to predictive accuracy when interpreting short-horizon market signals.

1. Introduction

Limit order books encode the interaction of liquidity supply and demand at the resolution where short-horizon prices form. Their evolution combines persistent order-flow patterns with rapidly changing depth, spreads and volatility. A forecasting model must extract weak directional information from this changing state while distinguishing useful temporal context from redundant observations. The resulting learning problem is sensitive to the observation window, the

prediction horizon and the market interval used for evaluation. These choices belong to the empirical specification of a model, alongside its architecture.

Predictive quality and signal quality are distinct empirical objects. Macro-F1 measures class-balanced discrimination; a directional signal additionally depends on which predictions are acted upon, how often they occur and how their interpretation changes under friction and delay. A higher-confidence subset can contain fewer directional observations, a different class mixture and a different distribution of future outcomes. Consequently, improving an aggregate prediction score need not improve a confidence-filtered diagnostic. Conversely, an apparently favourable signal statistic can emerge as the selected sample becomes extremely small.

We introduce ChronosLOB to study these relationships within a common temporal evaluation framework. The principal experiment is a 180-cell FI-2010 neural benchmark, with matched model-family comparisons over fold, seed, lookback and horizon. A separate 30-pair study evaluates market-state self-supervised pretraining, denoted SSL-v2, against its own supervised control. Feature-removal experiments examine dependence on snapshot-derived information. Finally, a separate prediction cohort connects confidence filtering to directional coverage, unit-payoff diagnostics, configured cost stress, latency and adverse selection. Figure 1 summarises the experimental logic.

The results are structured by interactions rather than a single ranking. The matrix transformer has the higher overall macro-F1 and MCC, yet increasing its lookback from 20 to 100 substantially reduces both measures. SSL-v2 improves average predictive and probabilistic metrics at horizon 10 but degrades them at horizon 50. Removing the snapshot order-flow proxy consistently weakens classical forecasts, whereas several level and imbalance features have much smaller marginal effects. Confidence filtering then changes the interpretation of a forecast again: at horizon 50, supervised macro-F1 initially rises with selection before falling as directional coverage contracts.

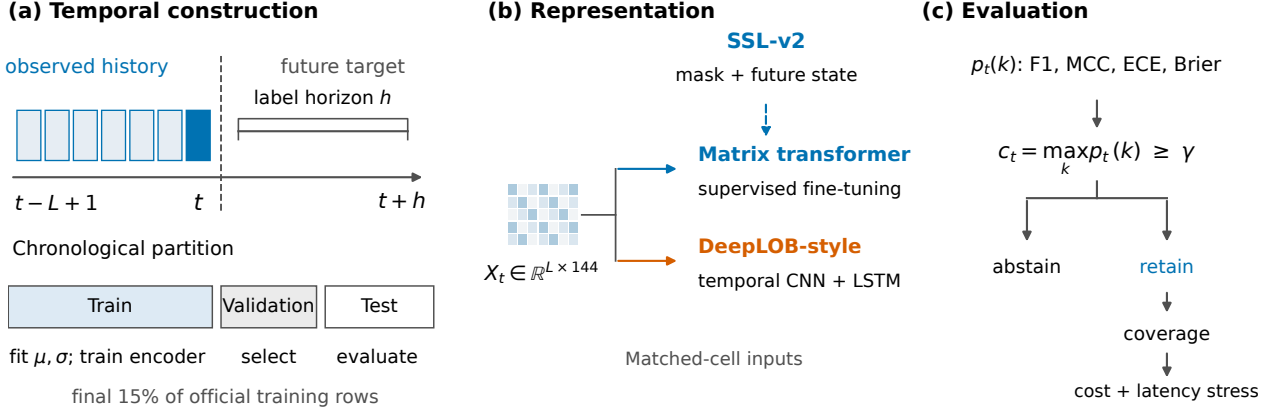


Figure 1. Experimental design. (a) A predictor receives a past-only window; its target describes a future horizon. The official training partition supplies both fitting rows and a final validation segment. (b) The matrix transformer and temporal CNN–LSTM share matrix inputs; SSL-v2 transfers a training-only pretrained encoder into supervised fine-tuning. (c) Raw probability assessment precedes confidence filtering, coverage measurement and execution-aware stress diagnostics. The benchmark, SSL and execution results use their respective cohorts.

These findings motivate three contributions:

- A controlled neural comparison that quantifies the interaction between model family, context length and prediction horizon across 180 temporally evaluated runs.
- A matched representation study and feature-stability analysis that identify horizon-dependent pretraining effects and persistent dependence on snapshot state changes.
- A joint evaluation of discrimination, probabilistic reliability, selective coverage and execution sensitivity that exposes the gap between forecast scores and directional-signal diagnostics.

2. Related Work

Limit order books aggregate individual participants’ price and size decisions into a structured, continuously updated record of displayed liquidity (Gould et al., 2013). FI-1010 provides normalised limit order book representations and an anchored chronological evaluation protocol for mid-price prediction (Ntakaris et al., 2018). DeepLOB combines convolutional processing with recurrent temporal modelling (Zhang et al., 2019), and large-scale deep learning has more broadly been shown to recover universal, cross-market structure in price formation (Sirignano & Cont, 2019). Our convolutional reference uses two temporal convolutions and an LSTM, giving a compact DeepLOB-inspired comparison under the same matrix-input protocol. The comparison concerns these implemented model families and their training conditions.

Order-flow imbalance connects changes in supply and demand to short-term price movements (Cont et al., 2014). This motivates investigating inter-snapshot changes alongside contemporaneous book geometry. We define the snapshot order-flow proxy through signed differences of observed price and size fields, and evaluate event-level features separately using replay. This distinction preserves the connection to microstructure while respecting the information available in each data representation.

Transformers provide a flexible mechanism for modelling sequence dependencies (Vaswani et al., 2017). Recent LOB work has explored attention over both temporal and feature dimensions. TLOB also examines horizon-dependent labelling and the relationship between classification and transaction-cost assumptions (Berti & Kasneci, 2025). The broader LOBCAST benchmark documents substantial changes in model performance across datasets and market conditions (Prata et al., 2024). These studies motivate reporting configuration sensitivity and evaluating market interpretation beyond an aggregate forecasting score.

Probability calibration measures agreement between stated confidence and empirical correctness; temperature scaling is an established post-hoc approach (Guo et al., 2017). We assess the raw neural probabilities with ECE and the multiclass Brier score (Brier, 1950). Confidence thresholding connects this assessment to selective prediction and its coverage trade-off (Geifman & El-Yaniv, 2019). Financial evaluation additionally requires controlling which observations can influence fitting and selection (López de Prado, 2018). ChronosLOB brings these strands together through matched experiments and explicit coverage and stress diagnostics, with each experimental cohort evaluated within its

own training regime.

3. Problem Formulation and Method

3.1. Book State and Prediction Target

Let b_t^j, a_t^j and $v_t^{b,j}, v_t^{a,j}$ denote bid and ask prices and displayed sizes at level j . The underlying mid-price is $m_t = (a_t^1 + b_t^1)/2$. The neural input is a sequence $X_t = (x_{t-L+1}, \dots, x_t) \in \mathbb{R}^{L \times 144}$ containing the 40 normalised price and size fields for ten levels and the 104 supplied features. All label columns and partition identifiers are excluded from the input.

The target $y_t^{(h)} \in \{\text{down}, \text{stationary}, \text{up}\}$ is the supplied FI-2010 annotation for horizon h . Its construction thresholds the relative displacement of a future average mid-price from the current mid-price (Ntakaris et al., 2018):

$$r_{t,h} = \frac{\bar{m}_{t,h}^+ - m_t}{m_t}, \quad \bar{m}_{t,h}^+ = \frac{1}{|\mathcal{H}_h(t)|} \sum_{u \in \mathcal{H}_h(t)} m_u, \quad (1)$$

where $\mathcal{H}_h(t)$ is the future interval specified by the benchmark. The experiments consume the released labels directly. We retain their horizon identifiers, and measure lookbacks and simulated latency in matrix rows. Thus the forecasting target remains fixed when the context length changes.

3.2. Temporal Evaluation

Each official fold supplies chronological training and test partitions. The final 15% of the official training rows forms the validation segment; earlier rows fit the model. Input windows are contiguous and contained within their assigned partition, and feature-wise standardisation is fitted only on the fitting rows before being applied to validation and test inputs. Checkpoints are selected by validation macro-F1, and the best checkpoint is restored before test evaluation. The released FI-2010 normalisation and labels are used as supplied.

SSL-v2 imposes the further condition that each pretraining input window and its future auxiliary target lie within the same partition; auxiliary bin boundaries use fitting rows only.

3.3. Forecasting Models

The matrix transformer projects each input row from 144 to 64 dimensions, adds learned positional embeddings and applies two encoder layers with four attention heads, GELU activation and a feed-forward width of 128. Mean pooling over the observed window feeds a three-class linear head. Attention spans the entire observed window, which ends at the prediction point.

The DeepLOB-style reference treats input features as channels. Two one-dimensional convolutions of kernel width 3 produce 32 channels, each followed by batch normalisation, ReLU and dropout. A single-layer LSTM with hidden size 64 processes the resulting sequence, and its final state feeds a three-class linear head. Both families use dropout 0.10, cross-entropy loss and Adam. Table 1 specifies the common training schedule; Appendix A gives the full architecture and objective settings.

3.4. Market-State Self-Supervision

SSL-v2 combines structured reconstruction with four future-state classification tasks. Feature groups are masked with probability 0.30 and reconstructed using mean squared error on masked entries. Auxiliary heads predict spread widening and three-bin targets for future volatility, mid-price change and imbalance. The latter bin edges are training-fitted terciles. With task cross-entropies $\mathcal{L}_s, \mathcal{L}_v, \mathcal{L}_r, \mathcal{L}_i$, the objective is

$$\mathcal{L}_{\text{SSL}} = \mathcal{L}_{\text{rec}} + \frac{1}{2}\mathcal{L}_s + \frac{1}{2}\mathcal{L}_v + \mathcal{L}_r + \frac{1}{2}\mathcal{L}_i. \quad (2)$$

Future auxiliary states use an offset of h matrix rows. Pretraining lasts five epochs, after which the input projection, positional embeddings and transformer encoder initialise the supervised model. All encoder parameters are fine-tuned.

We compare pretraining with random initialisation at identical fold, horizon, seed, lookback, architecture and preprocessing. The 30 pairs use batch size 1024; the 180-cell neural grid uses 256. Each SSL delta therefore refers to its matched supervised control. The two training regimes answer separate empirical questions.

3.5. Probabilistic Reliability and Selection

For logits z_t , let $p_t = \text{softmax}(z_t)$, $\hat{y}_t = \arg \max_k p_t(k)$ and $c_t = \max_k p_t(k)$. The reported neural probabilities are unscaled. ECE uses ten equal-width confidence bins; the Brier score averages the sum of squared errors over the three class probabilities. These measures assess calibration and overall probabilistic accuracy, respectively. No post-hoc temperature fit is applied to the reported neural results.

At confidence threshold γ , distinguish retained predictions from active directional predictions:

$$\begin{aligned} \mathcal{R}_\gamma &= \{t : c_t \geq \gamma\}, \\ \mathcal{A}_\gamma &= \{t \in \mathcal{R}_\gamma : \hat{y}_t \neq \text{stationary}\}, \quad A_\gamma = |\mathcal{A}_\gamma|/N. \end{aligned} \quad (3)$$

Selective macro-F1 is computed on $\mathcal{R}_\gamma$, whereas active coverage uses $\mathcal{A}_\gamma$. Reporting both separates confidence-based abstention from predictions of a stationary market. Thresholds are evaluated as a diagnostic sweep rather than selected to maximise a test statistic. A run group contributes to a

Table 1. Experimental protocol. Horizon values name the supplied FI-2010 labels; lookbacks count matrix rows. SSL-v2 uses its own matched supervised runs. Architecture details and auxiliary targets are expanded in Appendix A.

Component	Specification
Temporal construction	Official folds 1–5; validation is the final 15% of official training rows; training-fitted standardisation; contiguous partition-contained windows.
Neural comparison	90 runs per family; seeds 0/1/2; lookbacks 20/50/100; horizons 10/50; 144 input fields.
Matrix transformer	Model width 64; 4 heads; 2 layers; feed-forward width 128; learned positions; mean pooling.
DeepLOB-style reference	Two temporal convolutions, kernel 3, 32 channels, batch normalisation and ReLU; one LSTM layer, hidden size 64; final-state pooling.
Optimisation	Adam; learning rate 10^{-3} ; weight decay 10^{-4} ; gradient clipping at 1; dropout 0.10; at most 25 epochs; validation macro-F1 patience 5; best-checkpoint restoration.
SSL-v2 comparison	30 matched pairs; lookback 50; horizons 10/50; five pretraining epochs; group mask probability 0.30; batch 1024, versus 256 in the neural grid.
Feature and signal studies	2,520 classical ablation runs plus a supplementary nonlinear slice; separate execution cohort of 135 run groups, three objectives and horizons 10/20/50.

threshold’s aggregate only if $|\mathcal{R}_\gamma| \geq 3$. Groups retaining fewer than three predictions are skipped and excluded from both the threshold mean and the contributing-group count.

3.6. Execution-Aware Diagnostic Proxies

We define signal quality through offline, unit-payoff diagnostics. An active directional prediction receives +1 when its class is correct and −1 otherwise, including a stationary outcome. For a fee setting f in basis points and spread multiplier s , the configured score is

$$Q_\gamma(f, s) = \sum_{t \in \mathcal{A}_\gamma} [21\{\hat{y}_t = y_t\} - 1 - 10^{-4}f - 0.01s]. \quad (4)$$

Active directional predictions are treated as aggressive actions, so the spread term enters as a crossing penalty. The spread unit is a fixed 0.01 diagnostic penalty. Cost stress is the percentage reduction from the zero-cost score, normalised by its absolute value within each run and then averaged. In the selected threshold sweep, the stored adjusted score equals the gross score $Q_\gamma(0, 0)$; non-zero cost settings enter the separate stress comparison.

Latency pairs a signal at row t with the outcome label at row $t + \ell$, within the same partition, and reports the score change relative to $\ell = 0$. A signal-change rate measures transitions in the thresholded directional sequence. The adverse-selection proxy is the fraction of selected directional signals followed by an opposite-direction label; stationary outcomes are excluded from its adverse count. These definitions distinguish classification mistakes, signal activity and opposite-direction exposure.

4. Experimental Design

The empirical backbone is the public FI-2010 no-auction z-score representation, derived from five NASDAQ Nordic stocks over ten trading days (Ntakaris et al., 2018). We use official folds 1–5. The neural grid contains $2 \times 5 \times 3 \times 3 \times 2 = 180$ runs: two model families, five folds, seeds 0–2, lookbacks 20/50/100 and horizons 10/50. Training preserves sequence order and uses validation early stopping under the schedule in Table 1.

The SSL study contains 60 completed runs forming 30 matched pairs. Feature ablations use logistic and ridge classifiers across five folds, three seeds and horizons 10/20/50, with 12 snapshot-derived groups and six ablation modes. Their stability summaries additionally include ten gradient-boosting comparisons for four groups. We analyse matched remove-one-group differences; feature-group sample sizes are therefore 90 or 100.

The execution cohort contains 135 run groups across supervised, masked-reconstruction and next-field objectives and horizons 10/20/50. It is distinct from the validation-selected neural grid and the SSL-v2 cohort. Confidence thresholds range from 0.33 to 0.95; at each threshold, a run group contributes only if it retains at least three predictions after filtering, the eligibility rule defined in Section 3.5. The reported stress summaries provide two non-zero cost settings and a latency offset of ten rows. Predictive metrics include macro-F1, MCC, accuracy, class-level F1, ECE and Brier score. Aggregates weight runs equally; sample standard deviations describe variation across related experimental cells. Execution threshold summaries provide means and contributing-group counts.

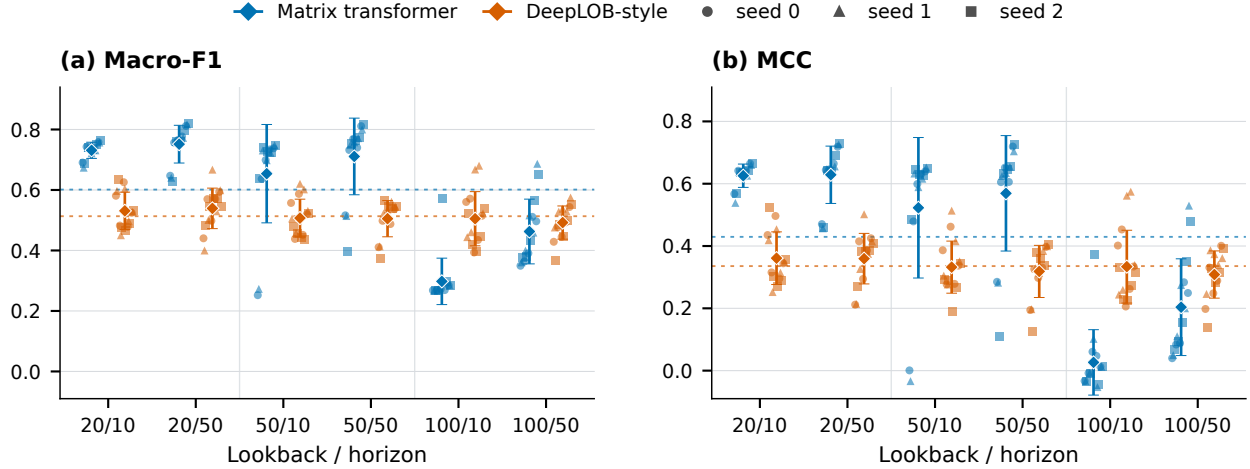


Figure 2. Neural benchmark stability across all 180 runs. Small points show individual fold and seed cells; marker shapes identify seeds. Diamonds and intervals denote the mean and one sample standard deviation over 15 cells per model/lookback/horizon condition. Dotted lines mark family-wide means. The transformer advantage at short lookbacks reverses at lookback 100. Intervals summarise dispersion across the 15 related cells in each condition.

Table 2. Headline results. Upper rows are test means $\pm$ sample SD over 90 neural runs per family. Lower rows are mean matched changes, SSL-v2 minus its supervised control. Higher macro-F1 and MCC and lower ECE and Brier are preferable. SSL-v2 changes are computed against matched supervised controls.

Model or comparison	n	Macro-F1	MCC	ECE	Brier
Matrix transformer	90	0.6013 ± 0.1950	0.4294 ± 0.2727	0.0617 ± 0.0469	0.4318 ± 0.1622
DeepLOB-style	90	0.5133 ± 0.0670	0.3356 ± 0.0883	0.2223 ± 0.1222	0.6644 ± 0.1653
SSL-v2, all pairs	30	+0.0112	+0.0259	-0.0030	-0.0235
SSL-v2, horizon 10	15	+0.0559	+0.0978	-0.0182	-0.0755
SSL-v2, horizon 50	15	-0.0335	-0.0460	+0.0122	+0.0284

5. Results

5.1. Neural Benchmark

The matrix transformer achieves higher overall discrimination and probabilistic accuracy (Table 2). Its mean macro-F1 is 0.6013 ± 0.1950 , compared with 0.5133 ± 0.0670 for the DeepLOB-style reference; mean MCC is 0.4294 ± 0.2727 versus 0.3356 ± 0.0883 . The corresponding mean ECE values are 0.0617 and 0.2223, and Brier scores are 0.4318 and 0.6644. The transformer’s stronger mean coexists with substantially greater variation across configurations.

Figure 2 reveals the dominant interaction. At lookback 20, the transformer reaches macro-F1 0.7311 at horizon 10 and 0.7515 at horizon 50. At lookback 100, these means fall to 0.2978 and 0.4627; the reference achieves 0.5050 and 0.4923 in the same conditions. The transformer’s MCC at lookback 100 and horizon 10 is only 0.0265, showing that the deterioration extends beyond class-balanced F1. Across matched cells, transformer wins are 30/30, 28/30 and 6/30 at lookbacks 20, 50 and 100, respectively.

The short-context advantage is also more stable. Transformer macro-F1 standard deviations at lookback 20 are 0.0267 and 0.0624 for horizons 10 and 50. At lookback 50, they increase to 0.1626 and 0.1267. An empirical sum-of-squares decomposition attributes 73.3% of the transformer’s total macro-F1 variation to differences between the six lookback/horizon means, against 6.0% for the reference. This is a descriptive decomposition of the observed grid: context and horizon structure much more of the transformer variation.

Variation also persists within conditions. Averaging over the remaining dimensions, transformer fold means range from 0.4694 to 0.6708, while reference fold means range from 0.4818 to 0.5429. Seed means are closer: 0.5842–0.6206 for the transformer and 0.4964–0.5363 for the reference. The longer-window result therefore identifies a configuration sensitivity of the implemented learner, with performance depending strongly on how context length interacts with the architecture. Appendix B reports the complete condition-level discrimination and calibration statistics.

(a) Matched representation effects

(b) Snapshot feature stability

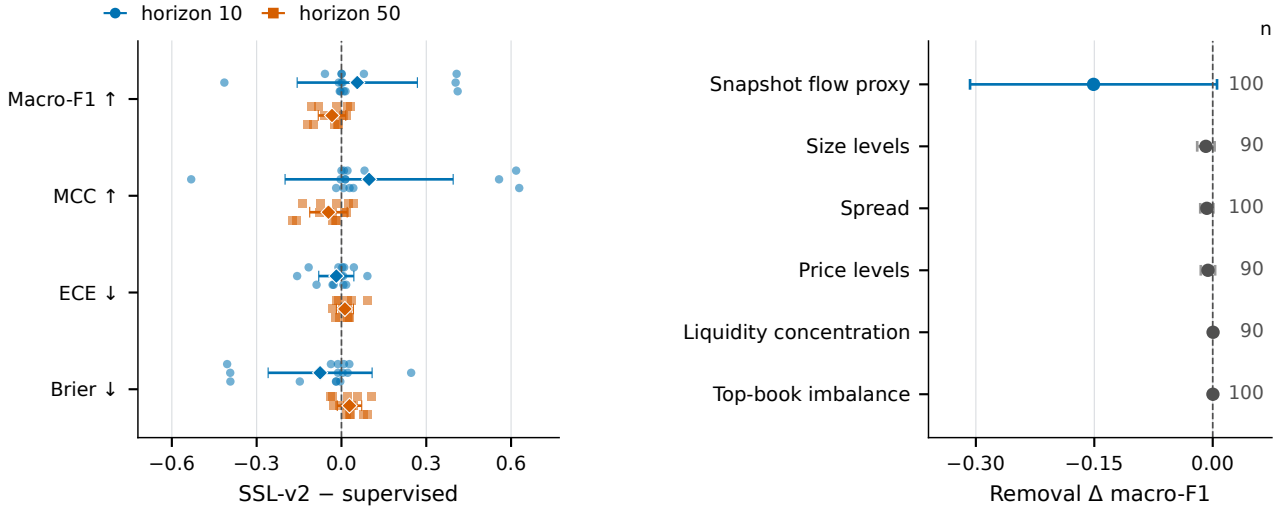


Figure 3. Representation and feature stability. (a) SSL-v2 minus supervised differences for all 30 matched cells. Diamonds and intervals show horizon means $\pm$ one sample SD; metric directions retain their original signs. Upward arrows identify higher-is-better metrics, downward arrows lower-is-better metrics. (b) Six feature-removal effects, with pooled means and sample SD reconstructed from horizon-level summaries. Counts identify the contributing comparisons. The snapshot flow proxy has the largest removal effect; all 12 groups appear in Appendix B.

5.2. Self-Supervised Representation Learning

SSL-v2 yields mean matched changes of $+0.0112$ macro-F1, $+0.0259$ MCC, -0.0030 ECE and -0.0235 Brier across 30 pairs. The signs of the means favour pretraining on all four measures. Individual outcomes are less uniform: macro-F1 improves in 13 pairs, and ECE decreases in 12. The positive mean therefore reflects the magnitude as well as the frequency of improvements.

Horizon separates the outcomes sharply (Figure 3a). At horizon 10, mean macro-F1 rises from 0.4280 to 0.4839, a change of $+0.0559$; MCC changes by $+0.0978$, ECE by -0.0182 and Brier by -0.0755 . Nine of 15 pairs improve macro-F1. At horizon 50, macro-F1 falls from 0.6699 to 0.6364, a change of -0.0335 ; the MCC, ECE and Brier changes are -0.0460 , $+0.0122$ and $+0.0284$. Only four of 15 pairs improve macro-F1.

The horizon-10 macro-F1 delta has sample SD 0.2126, compared with 0.0486 at horizon 50. Seed-level mean deltas are $+0.0277$, -0.0097 and $+0.0156$. Together, these results locate the benefit in a heterogeneous short-horizon regime. The separate reconstruction and next-field diagnostic comparisons show no consistent predictive improvement over supervision, motivating an objective tied more closely to future market state. The matched SSL-v2 experiment evaluates the combined market-state objective.

5.3. Feature Stability

We define the snapshot order-flow proxy as the vector of signed inter-snapshot differences in observed price and size fields, resetting differences at partition boundaries. Removing this group changes macro-F1 by -0.1510 and MCC by -0.1966 on average. Both metrics deteriorate in all 100 contributing comparisons. The model-specific macro-F1 changes are -0.2099 for logistic regression, -0.0507 for ridge and -0.3369 for the ten-run gradient-boosting slice.

The effect weakens with horizon: mean macro-F1 changes are -0.2397 , -0.1830 and -0.0347 at horizons 10, 20 and 50. Snapshot state changes are therefore especially influential for the shorter targets. Removing size levels, spread and price levels produces smaller mean changes of -0.0087 , -0.0075 and -0.0060 . Liquidity concentration and top-of-book imbalance removal yield changes near zero ($+0.0004$ and $+0.0002$).

The ablations quantify each group’s incremental contribution conditional on the remaining feature representation and the fitted classical model. Several groups encode overlapping aspects of book state, so a small removal effect can coexist with predictive information in the removed variables. The consistent snapshot-proxy effect identifies a stable source of incremental information under this ablation design, while its horizon dependence links feature stability to the representation results.

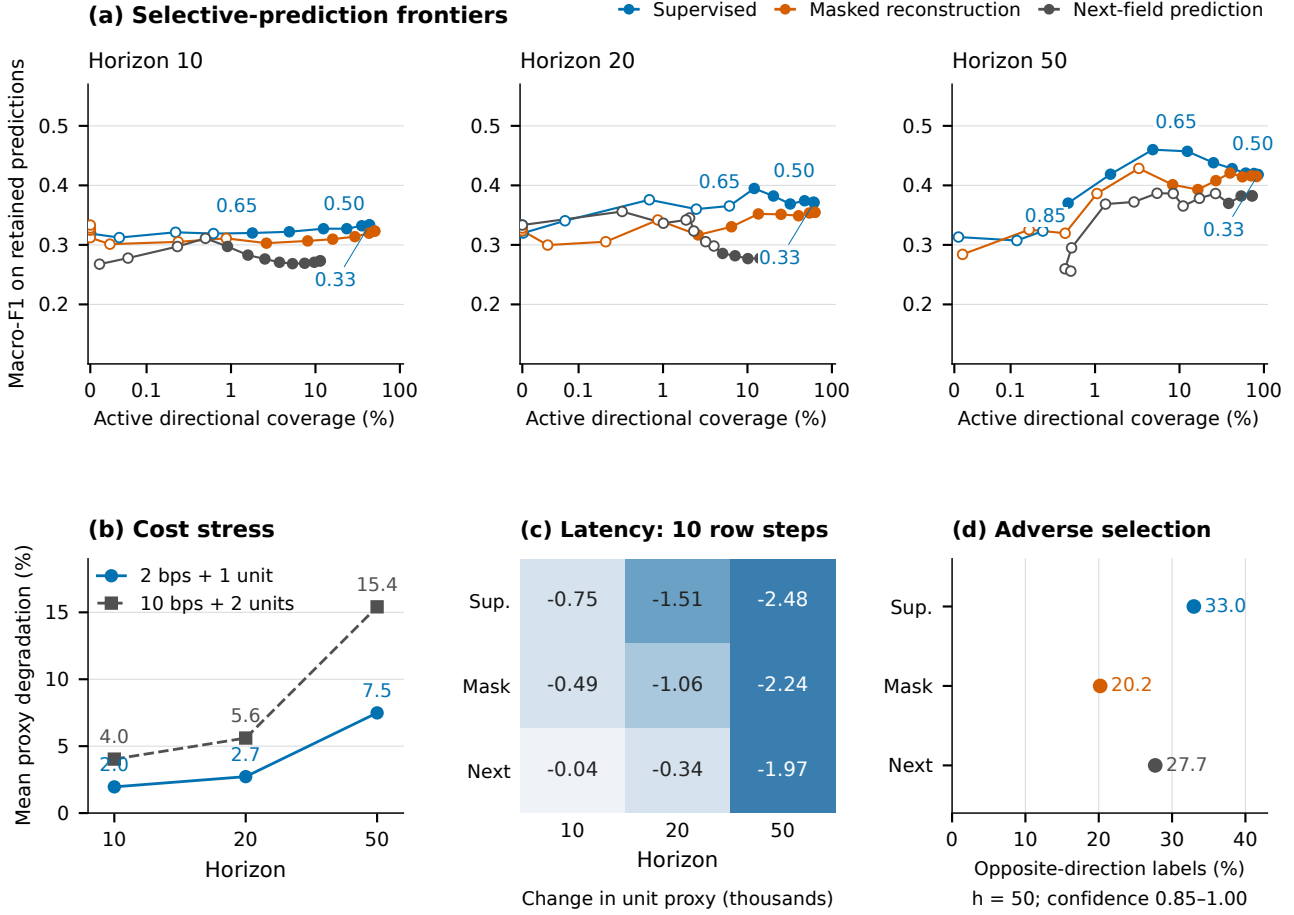


Figure 4. Forecasting-to-signal frontier in the separate 135-group diagnostic cohort. (a) Confidence-filtered macro-F1 against active directional coverage; annotations identify supervised confidence thresholds. Coverage is linear below 0.1% and logarithmic above it. Filled points average 15 groups; open points average fewer eligible groups. (b) Mean supervised cost degradation at the two recorded settings. (c) Change in the unit-payoff proxy at ten-row latency, in thousands. (d) Opposite-direction label fractions for horizon 50 in the 0.85–1.00 confidence bucket under the aggressive-crossing assumption. Threshold aggregates provide no dispersion estimates.

5.4. Forecasting Versus Signal Quality

Figure 4 joins selective prediction with cost, latency and adverse-selection diagnostics. The confidence–coverage curves vary by objective and horizon. For the supervised horizon-50 model, raising the threshold from 0.33 to 0.65 increases retained-sample macro-F1 from 0.4180 to 0.4600 while active directional coverage falls from 85.26% to 4.83%. Raising the threshold further to 0.70 reduces coverage to 1.52% and macro-F1 to 0.4187. Confidence selection initially improves discrimination but then changes the selected sample enough to reverse that gain.

At threshold 0.85, the supervised horizon-50 active fraction is 0.12%, and macro-F1 is 0.3073 over the ten groups that retained at least three predictions. Eligible-group counts also decline for the other objectives. This changes both the selected observations and the composition of the aggregate. Figure 4 marks those points explicitly, allowing the extreme-

confidence tail to be read together with its reduced support.

The distinction is visible even for nearly identical unfiltered forecasting scores. At horizon 50, supervised and masked-reconstruction macro-F1 are 0.4180 and 0.4148, with ECE 0.0733 and 0.0854. At threshold 0.70, their active fractions are 1.52% and 3.30%, while their mean zero-cost unit scores are -26.9 and -101.3 . Next-field prediction has macro-F1 0.3823 and ECE 0.0846, with 5.43% active coverage and a unit score of -219.8 at the same threshold. A larger active set exposes more observations to both correct and incorrect directional predictions. The movement of an aggregate unit score towards zero as confidence rises is therefore partly a contraction in activity, and must be assessed with the selected-sample prediction metric.

Configured frictions add a separate sensitivity. For supervised predictions, mean degradation under a two-basis-point fee and one spread unit rises from 1.96% at horizon 10 to

7.49% at horizon 50. Under ten basis points and two units, the corresponding values are 4.03% and 15.41%. These percentages average within-run reductions relative to each run’s absolute zero-cost score; they are not ratios of the plotted aggregate means.

Latency produces an additional change in diagnostic ordering. At ten rows, the supervised unit score changes by -748.2 , -1508.2 and -2476.7 at horizons 10, 20 and 50. Masked reconstruction and next-field prediction also deteriorate most at horizon 50 in the selected summaries. This larger absolute change depends on activity and the unit-payoff construction as well as forecast persistence. The latency diagnostic quantifies score sensitivity to a ten-row displacement within this evaluation protocol.

Finally, the high-confidence horizon-50 adverse-selection fractions are 32.97% for supervision, 20.19% for masked reconstruction and 27.70% for next-field prediction. The masked objective thus combines a similar unfiltered macro-F1 with a different opposite-direction exposure. These bucket-level comparisons complement the coverage frontier, enabling cross-objective comparison at this confidence bucket. The empirical object is a joint frontier over discrimination, coverage and sensitivity, with no single confidence threshold preferred by all measures.

6. Discussion and Scope

The neural benchmark establishes a strong interaction between architecture and context length: a higher family-wide mean can conceal a systematic reversal at longer lookbacks. SSL-v2 adds a horizon-dependent representation effect, while feature removal identifies snapshot state changes as a persistent source of incremental predictive information. Selective prediction alters class composition, activity and the set of contributing runs, so its interpretation requires

coverage and stress measurements alongside discrimination. FI-2010 supplies normalised snapshot representations and categorical future labels; the execution quantities here are offline diagnostic proxies. The execution layer evaluates sensitivity to confidence, configured penalties, latency and opposite-direction outcomes. Controlled event replay and aggregate-L2 reconstruction (Appendix C) provide complementary data-processing extensions through which such observations can enter the same evaluation design.

7. Conclusion

ChronosLOB shows that short-horizon LOB performance is governed by interactions among context length, prediction horizon and confidence selection. The matrix transformer leads at short lookbacks but reverses at the longest context; SSL-v2 improves the horizon-10 comparison while degrading horizon 50; and confidence filtering changes both retained predictive quality and active directional coverage. These effects are obscured when evaluation collapses across experimental dimensions.

Representation learning should be tested against matched controls at each horizon and training regime. Feature importance should be measured through stable conditional effects across folds and model families. Selective forecasts should be accompanied by their retained and active fractions, with explicit attention to the support of high-confidence summaries. Execution-aware diagnostics should then quantify sensitivity to clearly specified penalties and delays. Together, these requirements turn model comparison into an assessment of the conditions under which predictive information remains useful. Market-microstructure forecasting should be evaluated jointly through predictive quality, probabilistic reliability, selective coverage and execution sensitivity.

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A. Model and Training Details

Table 3 records the architecture used by the neural experiments. The supplied 144 feature fields are retained for both model families. Standardisation uses only the fitting segment of each official training partition; labels and split indicators are excluded from the feature matrix. Training, validation and test loaders preserve order, use stride-one windows and retain the last incomplete batch. The supervised objective is unweighted multiclass cross-entropy. Model initialisation and training use the configured Python, NumPy and PyTorch seeds, with deterministic PyTorch algorithms requested.

The released FI-2010 labels are used as supplied. At the internal fitting-validation boundary, a horizon- h label can draw future support from at most the final h fitting rows; for the neural grid this corresponds to $h \in \{10, 50\}$. No additional label-horizon purge is applied at this internal boundary.

Table 3. Exact model and objective specification. Matrix-transformer settings apply to both random initialisation and SSL-v2 fine-tuning. Convolutional kernels operate along time; features form the input channels.

Setting	Matrix transformer	DeepLOB-style CNN-LSTM
Input projection	Linear, $144 \rightarrow 64$	Temporal convolution, $144 \rightarrow 32$ channels
Sequence backbone	2 transformer encoder layers; 4 heads per layer	2 temporal Conv1d layers; kernel width 3; stride 1; padding 1
Hidden processing	Feed-forward $64 \rightarrow 128 \rightarrow 64$; GELU; layer normalisation	Batch normalisation and ReLU after each convolution
Position / recurrence	Learned $L \times 64$ positional embedding	1 LSTM layer; input 32; hidden 64; unidirectional
Pooling and output	Mean over L rows; linear $64 \rightarrow 3$	Final LSTM state; linear $64 \rightarrow 3$
Dropout	0.10 in encoder layers	0.10 after each convolutional activation
Lookbacks	20, 50, 100 in the neural grid; 50 in SSL-v2	20, 50, 100
Shared optimisation	Adam; learning rate 0.001; weight decay 0.0001; gradient norm clipping 1.0; maximum 25 epochs; validation macro-F1 early stopping with patience 5; restoration of the best checkpoint.	
Batch sizes	256 for the 180-cell grid; 1024 for both sides of the 30-pair SSL-v2 comparison.	
SSL-v2 pretraining	5 epochs; Adam with the same learning rate, weight decay and clipping; structured mask probability 0.30; replacement value 0; contrastive term disabled.	
Auxiliary heads	Spread widening: binary classification. Volatility, mid-price change and future imbalance: three classes using training-fitted terciles. Loss weights: reconstruction 1, spread 0.5, volatility 0.5, return 1, imbalance 0.5.	

SSL-v2 groups features by their column names into price/depth, spread/microprice, imbalance and remaining temporal-context fields, using the non-empty groups present in the input. Future auxiliary labels are derived from the normalised best bid and ask fields. Spread widening compares the spread at $t + h$ with its current value. The return target is the mid-price difference over h rows; volatility is the standard deviation of the intervening one-row mid-price changes; imbalance uses the future bid-ask size difference divided by their sum. These are state proxies on normalised inputs. The pretraining window builder requires both the complete lookback and the future endpoint to lie in its assigned partition. Three-bin boundaries are fitted from eligible training pairs and then held fixed.

In the 180-run grid, realised training lengths range from 6 to 25 epochs in both families; the median is 25. These are validation-selected, budget-limited comparisons. The SSL analysis checks pair identity over fold, horizon, seed, lookback and family; the selected per-run records also contain matching architecture and preprocessing hashes within all 30 pairs.

B. Additional Empirical Results

The variance decomposition in Section 5.1 divides between-condition variation, $\sum_c n_c (\bar{F}_c - \bar{F})^2$, by total variation, $\sum_i (F_i - \bar{F})^2$, over six lookback/horizon conditions with $n_c = 15$. Feature intervals pool within-horizon and between-horizon sums of squares. For snapshot-proxy removal, the horizon-10 logistic/ridge subset gives mean Δ macro-F1 -0.1957 with 30/30 degradations; the broader horizon-20/50 subset gives -0.1032 with 65/65 degradations, including five gradient-boosting comparisons.

Table 4. Full neural condition breakdown: mean $\pm$ sample SD across 15 fold/seed cells per row. L is lookback and h the supplied horizon identifier. The supplied labels follow the FI-2010 representation, which groups ten underlying events; matrix-row offsets are reported separately.

Family	L	h	Macro-F1	MCC	ECE	Brier
Transformer	20	10	0.7311 $\pm$ 0.0267	0.6257 $\pm$ 0.0374	0.0323 $\pm$ 0.0191	0.2786 $\pm$ 0.0377
Transformer	20	50	0.7515 $\pm$ 0.0624	0.6287 $\pm$ 0.0920	0.0584 $\pm$ 0.0357	0.3527 $\pm$ 0.0834
Transformer	50	10	0.6539 $\pm$ 0.1626	0.5228 $\pm$ 0.2255	0.0493 $\pm$ 0.0402	0.3411 $\pm$ 0.1302
Transformer	50	50	0.7109 $\pm$ 0.1267	0.5692 $\pm$ 0.1853	0.0550 $\pm$ 0.0440	0.3907 $\pm$ 0.1350
Transformer	100	10	0.2978 $\pm$ 0.0767	0.0265 $\pm$ 0.1050	0.1162 $\pm$ 0.0598	0.6151 $\pm$ 0.0640
Transformer	100	50	0.4627 $\pm$ 0.1070	0.2037 $\pm$ 0.1553	0.0589 $\pm$ 0.0296	0.6127 $\pm$ 0.0776
CNN-LSTM	20	10	0.5310 $\pm$ 0.0619	0.3610 $\pm$ 0.0845	0.1998 $\pm$ 0.1223	0.6217 $\pm$ 0.1735
CNN-LSTM	20	50	0.5392 $\pm$ 0.0670	0.3594 $\pm$ 0.0811	0.1760 $\pm$ 0.0846	0.6155 $\pm$ 0.0904
CNN-LSTM	50	10	0.5071 $\pm$ 0.0625	0.3322 $\pm$ 0.0836	0.2350 $\pm$ 0.1475	0.6767 $\pm$ 0.2056
CNN-LSTM	50	50	0.5051 $\pm$ 0.0599	0.3184 $\pm$ 0.0837	0.2048 $\pm$ 0.1027	0.6653 $\pm$ 0.0928
CNN-LSTM	100	10	0.5050 $\pm$ 0.0896	0.3338 $\pm$ 0.1168	0.2361 $\pm$ 0.1607	0.6595 $\pm$ 0.2394
CNN-LSTM	100	50	0.4923 $\pm$ 0.0549	0.3085 $\pm$ 0.0758	0.2822 $\pm$ 0.0857	0.7480 $\pm$ 0.1223

Table 5. All snapshot feature-removal effects. Deltas compare removal with the matched all-feature control. The SD is pooled from the horizon-level macro-F1 difference summaries. The final column counts macro-F1 degradations. Ten supplementary gradient-boosting comparisons augment four groups; all other groups contain 90 logistic/ridge comparisons.

Removed feature group	n	Mean Δ F1	SD Δ F1	Mean Δ MCC	Degradations
Snapshot order-flow proxy	100	-0.1510	0.1564	-0.1966	100/100
Size levels	90	-0.0087	0.0110	-0.0115	78/90
Spread	100	-0.0075	0.0083	-0.0045	84/100
Price levels	90	-0.0060	0.0091	-0.0067	63/90
Liquidity concentration	90	+0.0004	0.0013	+0.0004	33/90
Top-of-book imbalance	100	+0.0002	0.0009	+0.0003	18/100
Volatility proxy	90	+0.0001	0.0008	+0.0007	42/90
Depth imbalance	100	-0.0000	0.0008	+0.0001	43/100
Depth slope	90	+0.0000	0.0008	+0.0001	21/90
Microprice	90	+0.0000	0.0005	+0.0000	51/90
Top of book	90	+0.0000	0.0008	+0.0001	57/90
Midprice	90	-0.0000	0.0008	+0.0001	63/90

The unstratified threshold means provide a compact comparison with the horizon-specific frontiers. At thresholds 0.33, 0.50, 0.70, 0.85 and 0.95, mean active fractions are respectively 0.544020, 0.216257, 0.017206, 0.002096 and 0.001290. The corresponding mean zero-cost unit scores are -6159.015, -2011.658, -146.894, -17.968 and 11.750. The first four entries average nine objective/horizon summaries. At 0.95, only four summaries remain, containing five contributing run groups in total. The favourable terminal score thus accompanies both extremely low activity and a changed aggregate population.

C. Replay Extensions

The synthetic extension generates additions, cancellations and trades under seven controlled regimes: stable liquidity, buy pressure, high volatility, sell pressure, low liquidity, wide spread and cancellation shock. With seed 0 and 3,000 generated events per regime, the initial book and subsequent stream contain 21,010 events. Sampling every five events yields 4,202 snapshots; a twenty-snapshot future label horizon yields 4,182 aligned labels. The replay produces zero crossed snapshots, negative-depth events or sequence gaps. It records 35 empty-book snapshots, preserving depletion states as part of the controlled stress experiment.

The chronological synthetic benchmark uses 60% fitting rows, 20% validation and 20% test. Test macro-F1 is 0.2104 for the majority classifier, 0.3445 for logistic regression, 0.3539 for ridge and 0.4805 for gradient boosting. These results were regenerated from the current code with the stated defaults. They establish learnable structure in the simulator and exercise the event-to-feature-to-label path; they are separate from the FI-2010 observations.

Aggregate-L2 replay initialises a depth snapshot, applies absolute quantity updates at each price and checks update continuity

Table 6. Execution stress by objective and horizon. Cost columns are mean percentage degradations for $(f, s) = (2, 1)$ and $(10, 2)$. The latency column is the mean change in the unit-payoff score at ten rows relative to zero latency; hit rate is evaluated after that shift. Cost percentages average run-level ratios.

Objective	h	Cost (2, 1)	Cost (10, 2)	Latency change	Lagged hit rate
Masked reconstruction	10	1.78	3.66	-485.3	0.1990
Masked reconstruction	20	2.54	5.22	-1,055.5	0.2727
Masked reconstruction	50	6.39	13.16	-2,236.5	0.3764
Next field	10	1.66	3.42	-37.2	0.2004
Next field	20	3.66	7.54	-337.5	0.2826
Next field	50	6.92	14.25	-1,974.1	0.3669
Supervised	10	1.96	4.03	-748.2	0.2076
Supervised	20	2.72	5.61	-1,508.2	0.2751
Supervised	50	7.49	15.41	-2,476.7	0.3781

and book invariants. The Binance-shaped fixture contains three diff messages spanning update identifiers 101–115, all applied successfully. It yields three snapshots, final bid/ask depths of three/four levels and zero continuity, crossing or quantity issues. The fixture tests aggregate price-level reconstruction.

D. Reproducibility

The source is available at <https://github.com/anannyenaik/chronos-lob>. This manuscript uses repository commit 509ec6c3a14a635ba2dcbcd4dc95690289901529. Its 37 selected-result files are verified against the committed SHA-256 manifest. Quantitative figures and table rows are generated directly from those files; the accompanying source package includes the inputs, figure script and unchanged ICML style files.

After installing the repository with `python -m pip install -e ".[torch,plots]"`, verify the selected summaries or regenerate its public figures with:

```
python scripts/plot_results.py --verify-only
python scripts/plot_results.py
```

The principal neural experiments are exposed through `run-fi2010-neural-proper-training-subset` and `run-fi2010-ssl-v2-benchmark`; their CLI help records the complete grid and training options. Additional entry points reproduce the feature-ablation, synthetic-replay and aggregate-L2 studies. Manuscript figures are regenerated with `python build.figures.py`, and the paper is compiled with `latexmk -pdf -jobname=ChronosLOB main.tex`.